Practical #4

Practical Title:
To apply GitHub-based collaboration techniques including merging and remote repository management

Theory:

1. What is Merging in Git?
Merging in Git is the process of **combining changes from one branch into another branch**. It is commonly used to integrate feature branches into the main branch.

2. Types of Merge:

1. Fast-forward Merge:

- Happens when there is a straight path (no divergence).
- Git simply moves the pointer forward.
- No extra merge commit is created.

2. Three-way Merge:

- Happens when branches have diverged.
- Git uses a common ancestor to merge changes.
- Creates a new merge commit.

3. What is Remote Repository?
A remote repository is a version of your project hosted on the internet or network, usually on platforms like GitHub.
It allows multiple developers to **access, collaborate, and manage code remotely**.

4. Common Remote Commands:
git remote
git push
git pull
git fetch

Procedure:

Step 1: Create Local Repository

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~
$ mkdir devops-practical4
cd devops-practical4
git init
Initialized empty Git repository in C:/Users/ACER/devops-practical4/.git/
```

Step 2: Add Remote Repository

```
ACER@DESKTOP-819C8K7 MINW64~/devops-practical4 (master)
$ git remote add origin
$ git push -u origin htttps://github.com/shreyabansall1
```

Step 3: Create Branch and Commit Changes

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4 (master)
$ git checkout -b feature1
echo "Feature work" > file.txt
git add .
git commit -m "Feature commit"
Switched to a new branch 'feature1'
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
[feature1 (root-commit) 5066180] Feature commit
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
```

Step 4: Push Branch to Remote

```
ACER@DESKTOP-819C8K7 MINGW64 ~/devops-practical4 (feature1)
$ git push -u origin feature1
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 221 bytetes | 221.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shreyabansalll/devops-practical4.git
* [new branch]  'fature1 -> feature1
Branch 'feature1' set up to track 'origin/feature1'.
```

Step 5: Merge Branch into Main

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4 (feature1)
$ git checkout -b main
git merge feature1
Switched to a new branch 'main'
Already up to date.
```

Step 6: Push Merged Code

```
ACER@DESKTOP-819C8K7 MINGW64 ~/devops-practical4 (mair
$ git push origin main
total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from
remote:Create a pull request for "main" on GitHub by visitin
To https://github.com/shreyabansalll/devops-practical4.git
* [new branch] main -> main
```

Step 7: Fetch and Pull Updates

```
ACER@DESKTOP-819C8K7 MINGW64 ~/devops-practical4 (main)
$ git fetch origin
$ git pull origin main
From https://github.com/shreyabansalll/devops-practical4
* branch main -> FETCH_HEAD
Already up to date.
```

Output Screenshots:
(Paste screenshots of all steps)

Activity / Task:

Task 1:
Create a remote repository and link it with local repository.

```
ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4 (main)
$ mkdir devops-practical4
cd devops-practical4
git init
Initialized empty Git repository in C:/Users/ACER/devops-practical4/devops-practical4/.git/

ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (master)
$ echo "Hello Practical 4" > file.txt

ACER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (master)
$ git add .
git commit -m "Initial commit"
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git touches it
[master (root-commit) 0d5c76f] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt
```

Repository Link:

Task 2:
Perform branch merging and push changes to GitHub.

```
CER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (main|MERGING)
$ git checkout -b feature2
witched to a new branch 'feature2'

CER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (feature2)
$ echo "New feature added" >> file.txt

CER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (feature2)
$ git add .
it commit -m "Feature2 update"
arning: in the working copy of 'file.txt', LF will be replaced by CRLF the next time Git
feature2 fde0cd9] Feature2 update
1 file changed, 5 insertions(+)

CER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (feature2)
$ git push origin feature2
numerating objects: 6, done.
ounting objects: 100% (6/6), done.
elta compression using up to 4 threads
ompressing objects: 100% (3/3), done.
riting objects: 100% (6/6), 539 bytes | 539.00 KiB/s, done.
otal 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
emote:
emote: Create a pull request for 'feature2' on GitHub by visiting:
emote:      https://github.com/rashipande/devops-practical4/pull/new/feature2
emote:
o https://github.com/rashipande/devops-practical4.git
* [new branch]      feature2 -> feature2

CER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (feature2)
$ git checkout main
witched to branch 'main'

CER@DESKTOP-8J9C8K7 MINGW64 ~/devops-practical4/devops-practical4 (main)
$ git merge feature2
pdating 0d5c76f..fde0cd9
ast-forward
file.txt | 5 +++++
1 file changed, 5 insertions(+)
```

Commands Used:

git branch feature

git checkout feature

git add .

git commit -m "Feature added"

git push origin feature

```
git checkout main

git merge feature

git push origin main
```

Task 3:
Explain the difference between git fetch and git pull.
Difference between git fetch and git pull:

git fetch	git pull
Downloads changes only	Downloads + merges changes
Does not modify working directory	Updates working directory
Safer for review	Direct update
Manual merge required	Automatic merge

Advantages:

1. Easy collaboration using remote repositories
2. Efficient branch management
3. Backup of code on remote servers
4. Supports team development
5. Tracks complete project history

Challenges:

1. Merge conflicts
2. Requires understanding of commands
3. Network dependency for remote operations